A few notes...

- **You will have one project (possibly 2) this semester involving NASTRAN/FEMAP (Chapter 7 in textbook)**
- **Project assignment (basically a HW) is to model the scenario of Problems 1.1 and 1.2 from the textbook and compare the results with your previously computed calculations (HW3); will likely be due in a couple of weeks**
- **On Mon Mar 30 (next class period), our "NASTRAN GTA" Anthony Seholm will come and provide a NASTRAN/FEMAP tutorial geared toward the project**
- **If possible, download and install NASTRAN/FEMAP on your own device, following Anthony's instructions that I provided last time (Wed Mar 25 module of Canvas), so you can follow along during his tutorial**

Example 4.5.1

- **The *y* coordinate of Point C is** $y_C = 0.5'' - 0.032'' = 0.468''$

- **We can use the equation of the NA from Part (a) to find the *z* coordinate of Point C →** $y = (1.315)z$ → $z_C = 0.468/1.315 = 0.3559''$

- **Also:**

$$A' = (0.064'')\overbrace{\left(\frac{9}{16}'' - 0.032'' - 0.3559''\right)}^{0.1743''}$$

$$\boxed{\bar{y}' = 0.468''}$$

$$\boxed{A' = 0.01117\ in^2}$$

$$\bar{z}' = 0.3559'' + \frac{1}{2}(0.1743'')$$

$$\boxed{\bar{z}' = 0.4431''}$$

Example 4.5.1

- **Inserting these values into** $q_{out} - q_{in} = \left[(12,230)\,\overline{y}' - (16,070)\,\overline{z}'\right]A'$ **yields:**

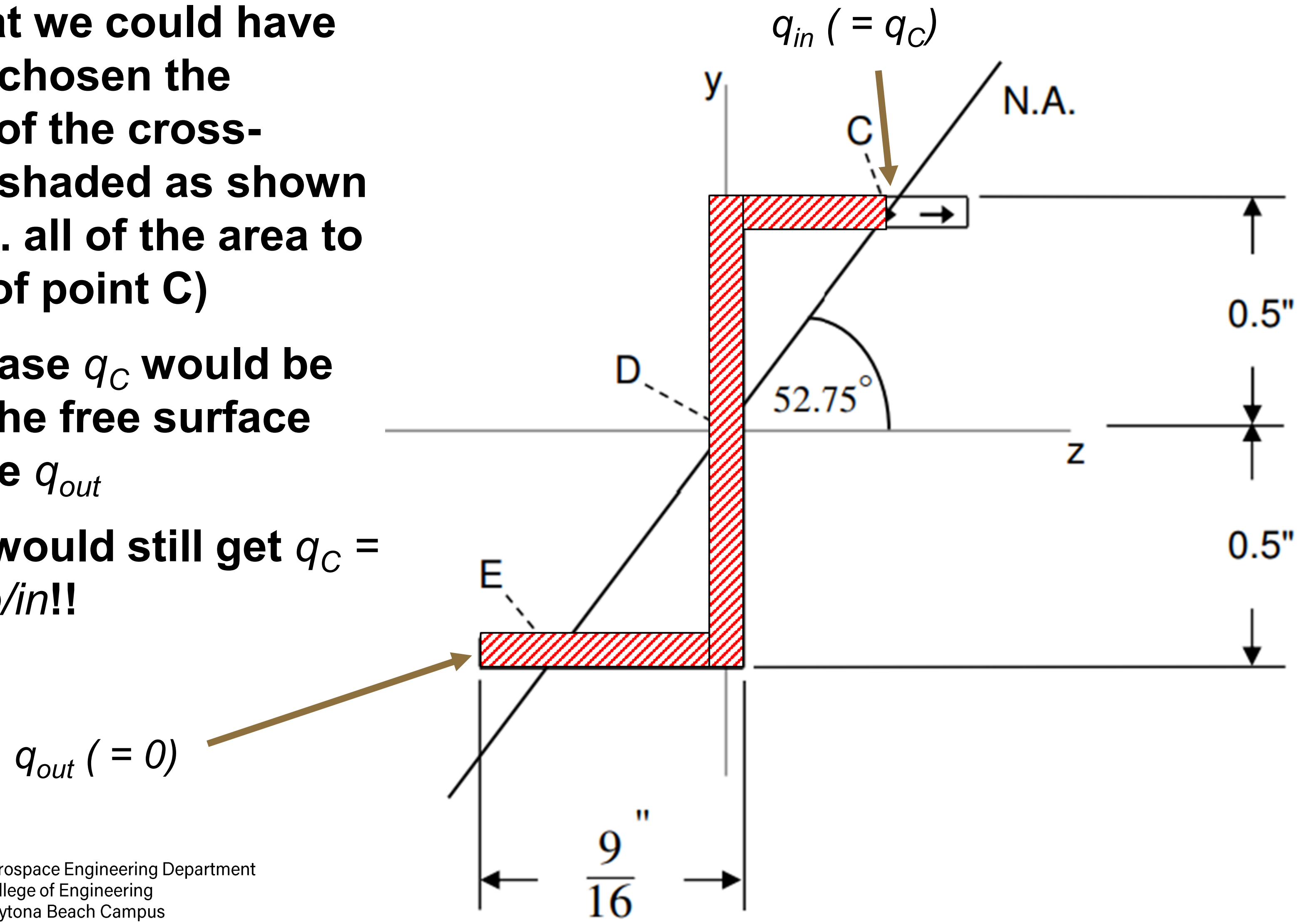
$$\Rightarrow \quad q_C - 0 = \left[(12,230)(0.468") - (16,070)(0.4431")\right]\left(0.01117\ in^2\right) = -15.60$$

$$\Rightarrow \quad \boxed{q_C = -\ 15.60\ \frac{lb}{in}}$$

My signs for q_C, q_D, and q_E differ from the textbook!

Example 4.5.1

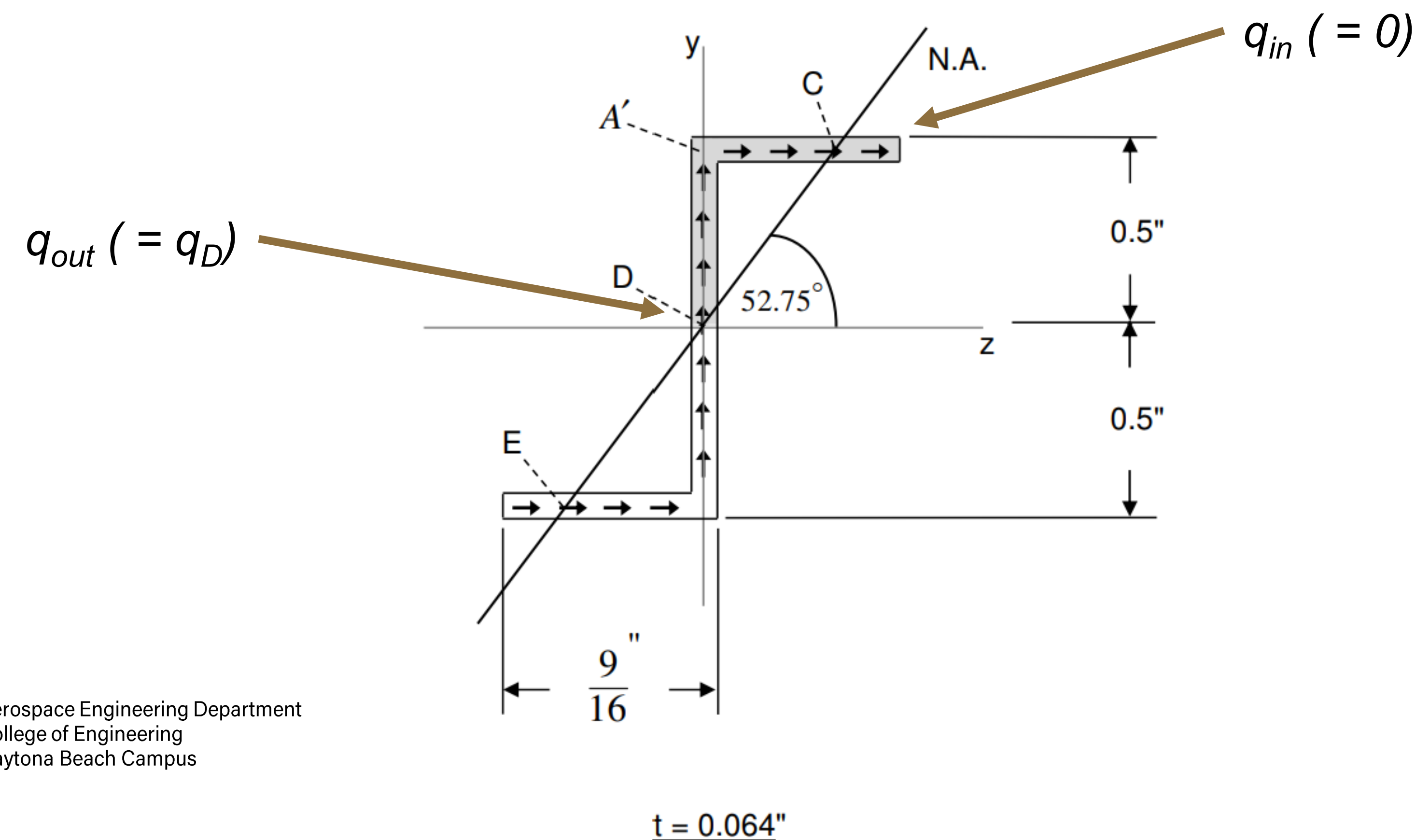
- **Note that we could have instead chosen the portion of the cross-section shaded as shown here (i.e. all of the area to the left of point C)**
- **In this case q_C would be q_{in} and the free surface would be q_{out}**
- **But we would still get q_C = *-15.60 lb/in*!!**



Example 4.5.1

- **Next we consider point D**

- **If we choose the portion shaded in gray below (i.e. all of the area above and to the right of point D), q_{out} will be q_D and q_{in} will be 0**



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Example 4.5.1

- **Point D is at the centroid of the cross-section, so *$y_D = z_D = 0$***

- **We also have:**

$$A' = (0.5")(0.064") + \left[\overbrace{\left(\frac{9}{16}" - 0.064"\right)}^{0.4985"}\right](0.064")$$

$$\boxed{A' = 0.06390\ in^2}$$

$$\bar{y}' = \frac{\bar{y}_1' A_1' + \bar{y}_2' A_2'}{A'}$$

$$\Rightarrow \quad \bar{y}' = \frac{[0.25"]\left[(0.5")(0.064")\right] + \left[\overbrace{0.5" - 0.032"}^{0.468"}\right]\left[(0.4985")(0.064")\right]}{0.06390\ in^2}$$

$$\Rightarrow \quad \boxed{\bar{y}' = 0.3589"}$$

$$\bar{z}' = \frac{\bar{z}_1' A_1' + \bar{z}_2' A_2'}{A'}$$

$$\Rightarrow \quad \bar{z}' = \frac{[0]\left[(0.5")(0.064")\right] + \left[\frac{0.4985"}{2} + 0.032"\right]\left[(0.4985")(0.064")\right]}{0.06390\ in^2}$$

$$\Rightarrow \quad \boxed{\bar{z}' = 0.008973"}$$

Example 4.5.1

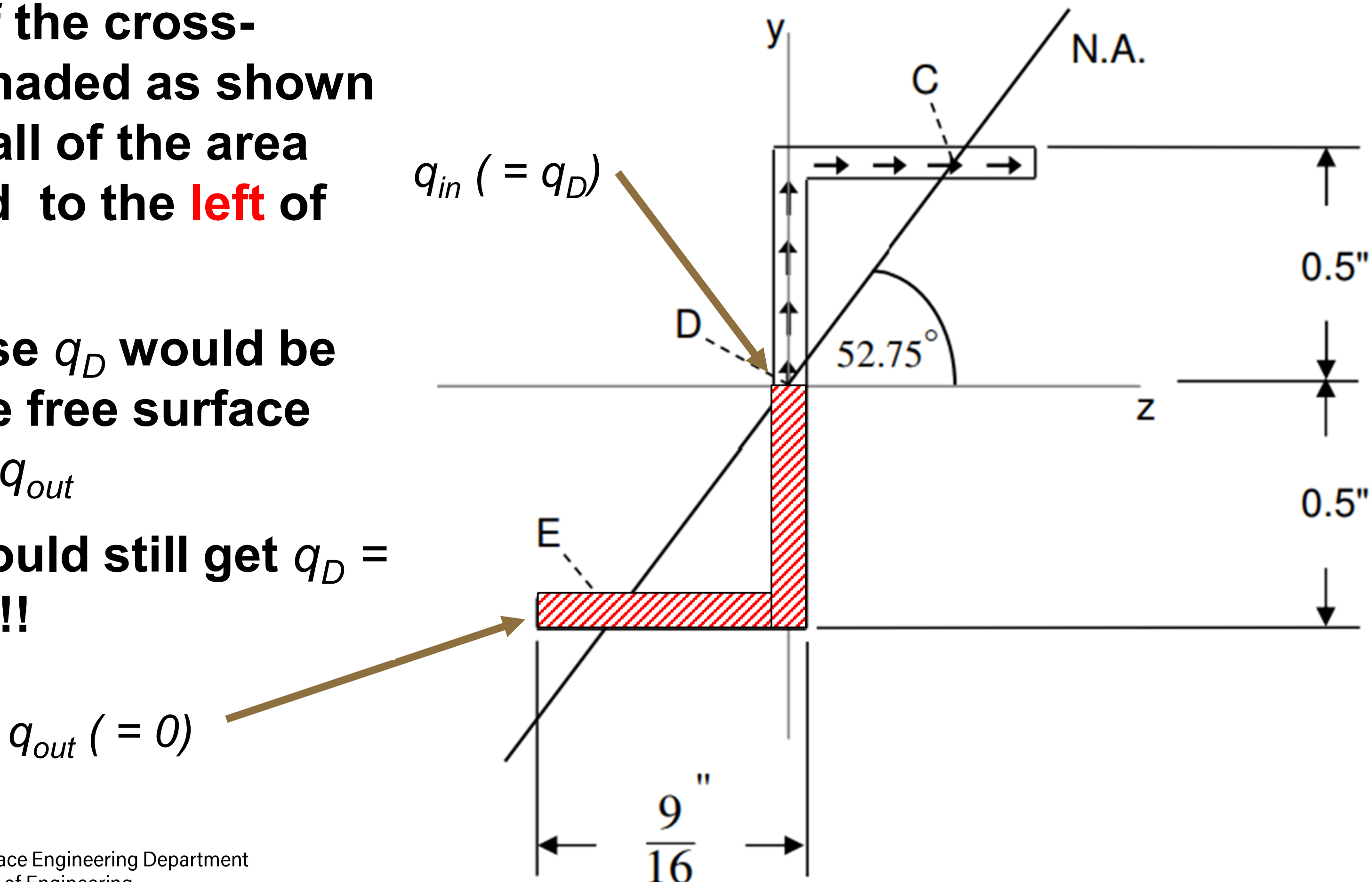
- **Inserting these values into** $q_{out} - q_{in} = \left[(12,230)\bar{y}' - (16,070)\bar{z}'\right]A'$ **yields:**

$$q_D - 0 = \left[(12,230)(0.3589") - (16,070)(0.008973")\right]\left(0.06390\ in^2\right) = 271.3$$

$$\Rightarrow \quad \boxed{q_D = 271.3\ \frac{lb}{in}}$$

Example 4.5.1

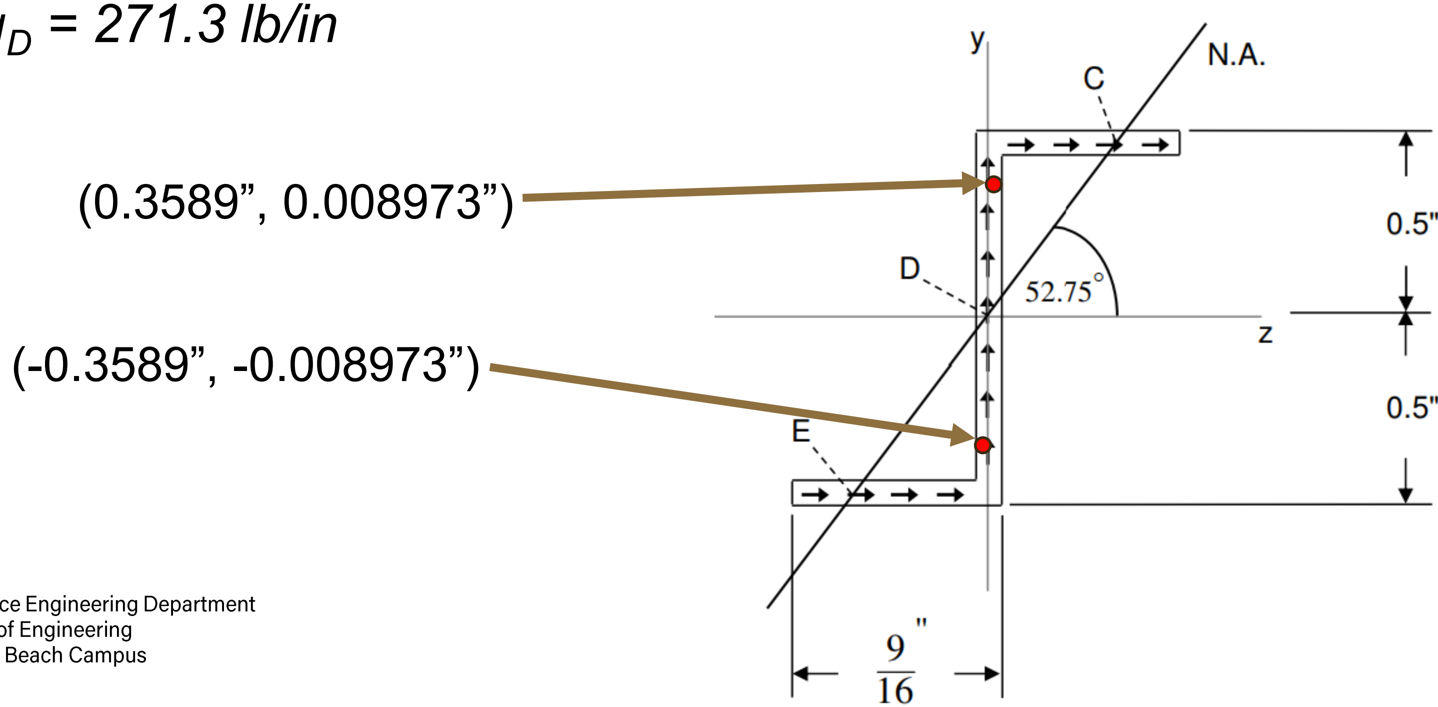
- **Again, we could have instead chosen the portion of the cross-section shaded as shown here (i.e. all of the area below and to the left of point D)**
- **In this case q_D would be q_{in} and the free surface would be q_{out}**
- **But we would still get q_D = *271.3 lb/in*!!**



Example 4.5.1

- **This is easy to show; due to symmetry (or “anti-symmetry”, as it were), if the centroid location of the original shaded area is (0.3589”, 0.008973”), then the centroid location of the alternate shaded area is (-0.3589”, -0.008973”)**

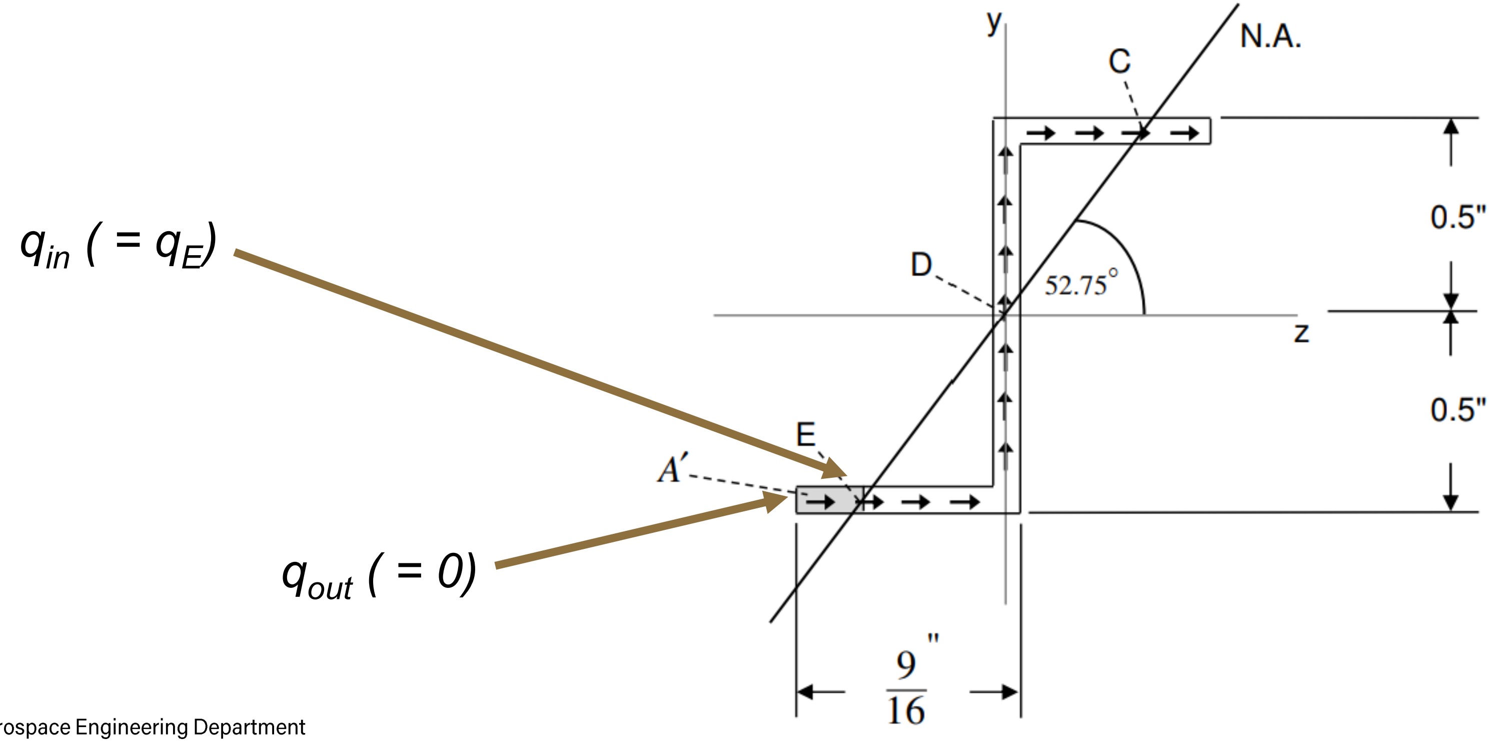
- **Then** $q_{out} - q_{in} = 0 - q_D = [12230(-0.3589) - 16070(-0.008973)]*(0.0639) = -271.3 \rightarrow q_D = 271.3\ lb/in$



Example 4.5.1

- **Finally we consider point E**

- **If we choose the portion shaded in gray below (i.e. all of the area to the left of point E), q_{in} will be q_E and q_{out} will be 0**



t = 0.064"

Example 4.5.1

- **The y coordinate of Point C is y_C = -(0.5" – 0.032") = -0.468"**

- **We can use the equation of the NA from Part (a) to find the z coordinate of Point C → $y = (1.315)z$ → z_C = -0.468/1.315 = -0.3559"**

- **Also:**

$$A' = (0.064")\overbrace{\left(\frac{9}{16}"-0.032"-0.3559"\right)}^{0.1743"}$$

$$\boxed{A' = 0.01117\ in^2}$$

$$\boxed{\bar{y}' = -0.468"}$$

$$\bar{z}' = -\left[0.3559"+\frac{1}{2}(0.1743")\right]$$

$$\boxed{\bar{z}' = -0.4431"}$$

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Example 4.5.1

- **Inserting these values into** $q_{out} - q_{in} = \left[(12,230)\bar{y}' - (16,070)\bar{z}'\right]A'$ **yields:**

$$\Rightarrow \quad 0 - q_E = \left[(12,230)(-0.468") - (16,070)(-0.4431")\right]\left(0.01117\ in^2\right) \quad = 15.60$$

$$\Rightarrow \quad \boxed{q_E = -15.60\ \frac{lb}{in}}$$

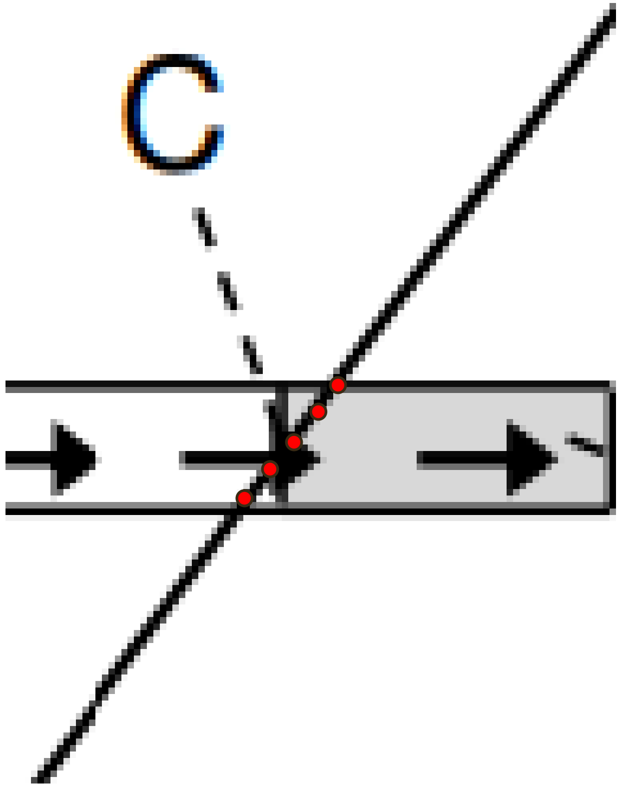
- **Again, we could have instead chosen all of the area above and to the right of point E (**q_{out} **would be** q_E **and** q_{in} **would be 0), but we would still get** $q_E = -15.60\ lb/in$

- **Therefore the largest shear flow is** *271.3 lb/in* **occurring at point D**

*******DONE*******

Example 4.5.1

- **Note that this solution involves the following assumption:**
- **When we look at point C, we see that the neutral axis does not actually intersect the cross-section at just one point here, but rather at a continuous locus of points (several of which are shown here in red)**
- **The same is true at points D and E; So where actually is the location of max shear flow?**

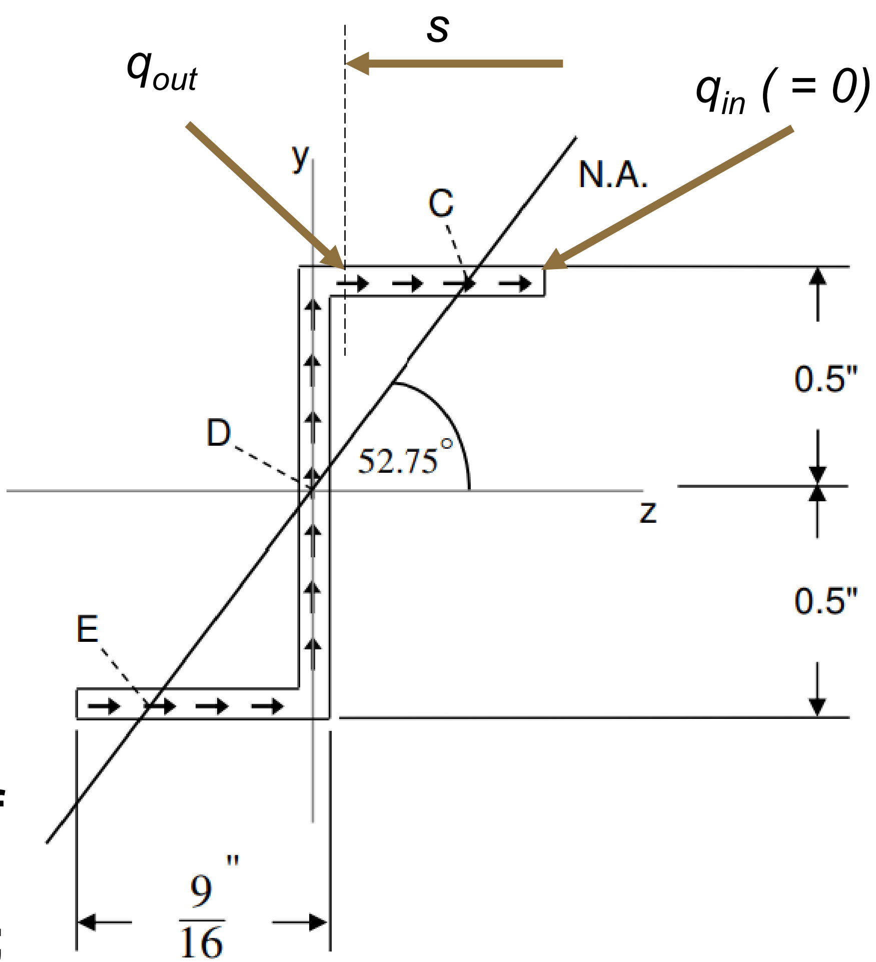


Example 4.5.1

- **It would be possible (but somewhat tedious) to work out the shear flow as a function of the length along the cross-section (call it *s*)**

- **To do this we could start from the upper right corner of the cross-section where $q_{in} = 0$, and choose a portion of the cross-section that gets larger as *s* increases**

- **We could then calculate the value of q_{out} (i.e. q) as a function of *s* throughout the entire cross-section; this would allow us to solve mathematically for the critical shear point**



Example 4.5.1

- **But without having to do all of that, here's what we can readily discern about *q* as *s* varies:**
  - *q* initially decreases from *0* (= *-15.60* at point C)
  - *q* then increases and at some point reaches *0* when $\bar{y} = (16070/12230)\bar{z} = 1.315\bar{z}$ (i.e. when the centroid is at point C)
  - *q* remains positive (= *271.3* at point D) and at some point decreases, reaching *0* when $\bar{y} = 1.315\bar{z}$ (i.e. when the centroid is at point E)
  - *q* continues negative and at some point increases, reaching *0* at the lower left corner of the cross-section
- **From this logic, we can probably conclude that *q* reaches a local min (= *-15.60*) at points C and E and a local max (= *271.3*) at point D**
- **So Radosta's assumption is likely a safe one, and max shear flow is in fact *271.3 lb/in*!**